

EFFICIENT TRADING FRONTIER

The efficient trading frontier (ETF) is the set of all optimal trading strategies. These are the strategies that contain the least risk for a specified cost and have the lowest cost for a specified risk. A rational investor is someone who will only trade via an optimal trading strategy. If an investor is trading via a strategy that is not on the efficient trading frontier it is unlikely that they will achieve best execution regardless of their actual execution costs.

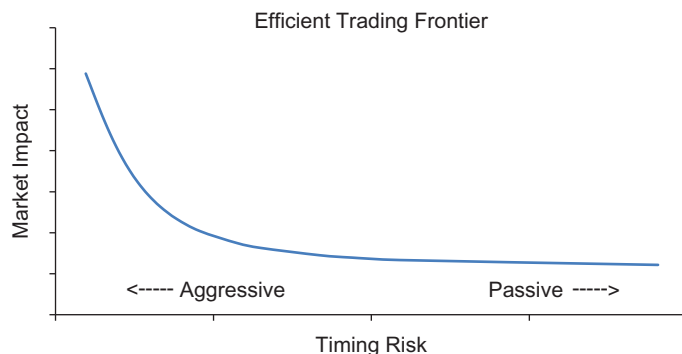
If a strategy is not optimal (e.g., it is above the ETF) then there exists a strategy with either (1) a lower cost for the same level of risk, (2) less risk for the same cost, or (3) a strategy with lower cost and less risk.

The efficient trading frontier is constructed via an optimization process. The general equation is:

$$\text{Min } L = \text{Cost} + \lambda \cdot \text{Risk} \quad (7.56)$$

where cost represents both market impact and alpha cost. In situations where investors do not have an alpha forecast or believe the natural price drift over the trading horizon to be zero they will only include the market impact cost component in the optimization.

Analysts then solve this equation for all values of $\lambda > 0$ and plot the sets of cost and risk. An example of the efficient trading frontier is shown in Figure 7.6. This figure illustrates the trade-off between market impact and timing risk. As the strategy becomes more aggressive timing risk decreases but market impact increases. As the strategy becomes more passive timing risk increases but market impact decreases. Market impact and timing risk are conflicting terms. Decreasing one term results in an increase in the other term. Unfortunately there is no way to simultaneously minimize both terms.



■ Figure 7.6 Efficient Trading Frontier.